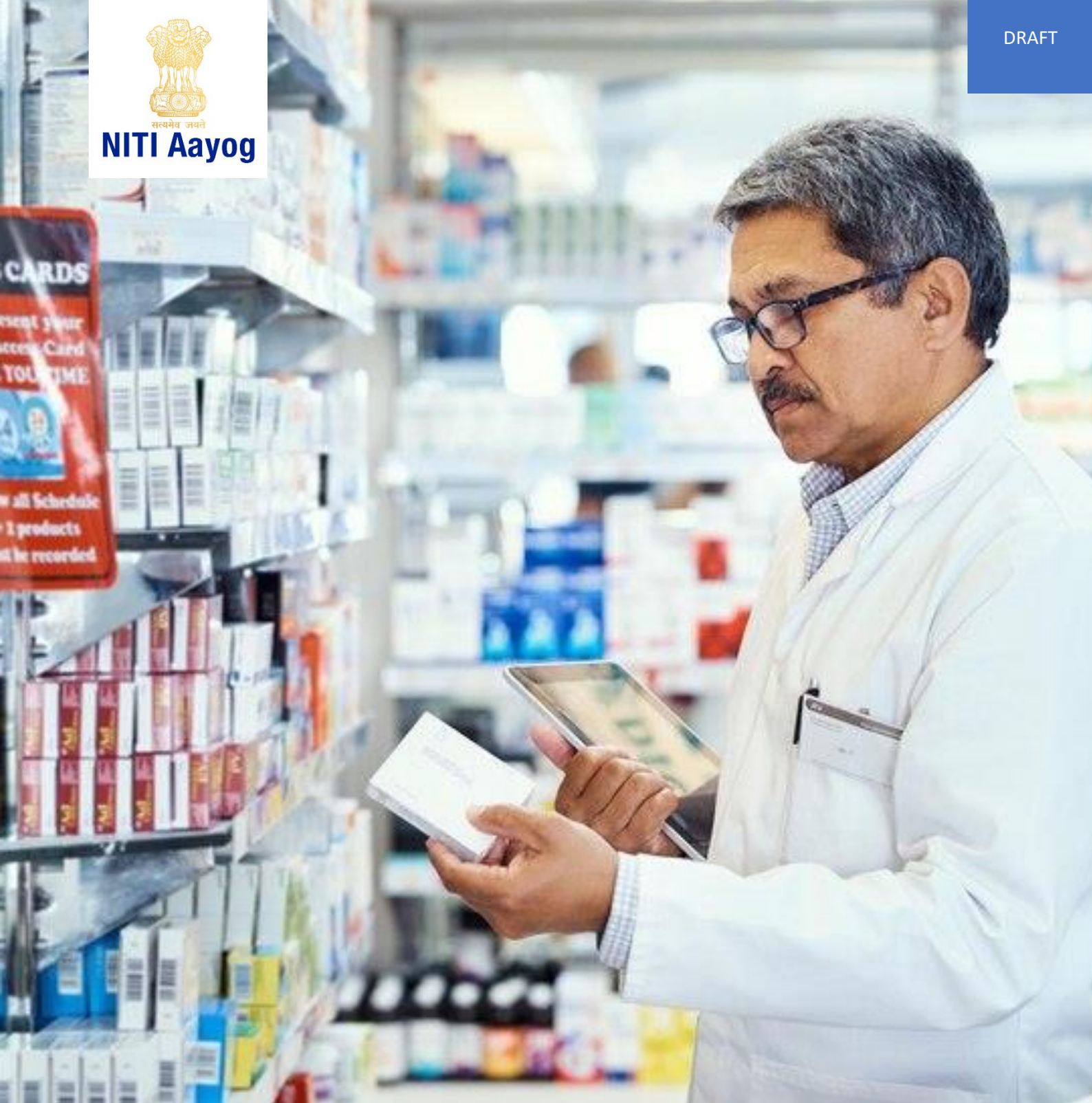




NITI Aayog

DRAFT



SECTOR AT A GLANCE

# Pharmaceuticals

Health Sector in India

DRAFT

Sector at a Glance

# Pharmaceuticals

Health Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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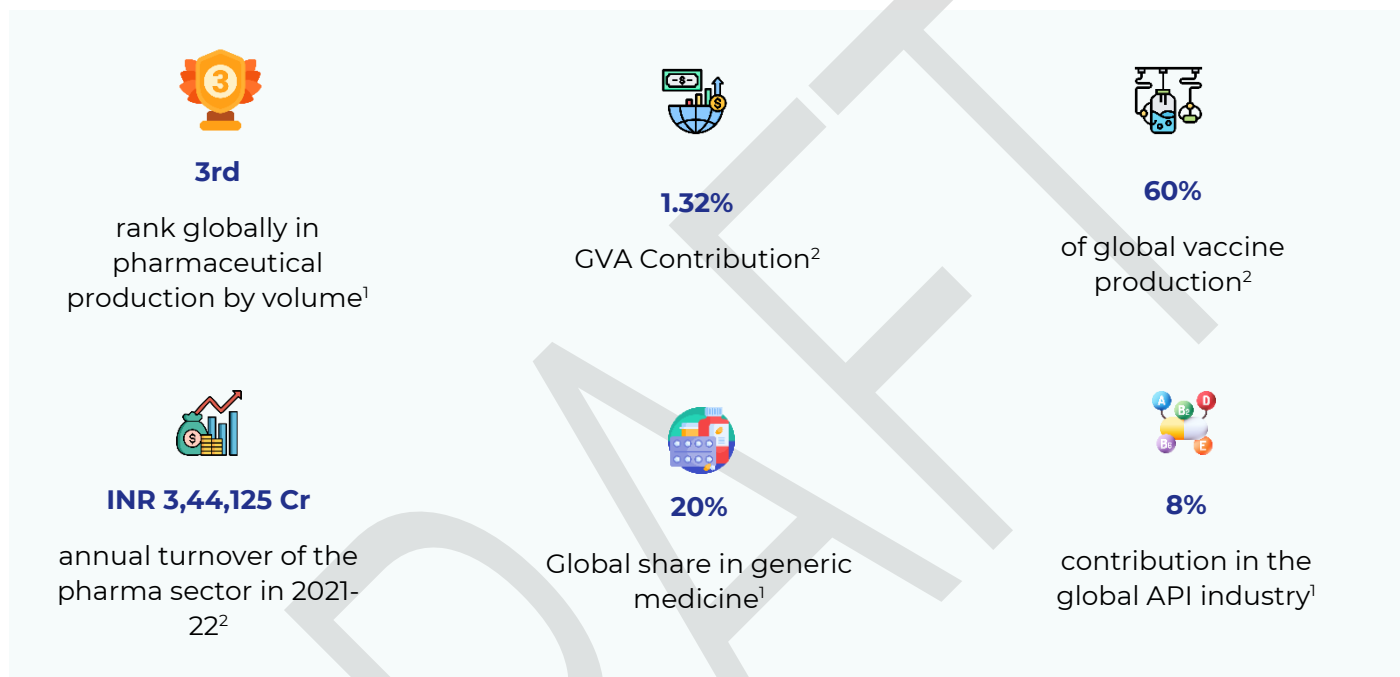
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# Overview

The Indian pharmaceutical industry plays a significant role globally, supplying affordable and low-cost generic drugs to millions of people worldwide thus impacting health outcomes on a global scale.<sup>1</sup> The Indian pharmaceutical industry **ranks third globally** in terms of volume and fourteenth in terms of value.<sup>2</sup> India is the only country that has the highest number of US Food and Drug Administration (USFDA) approved pharmaceutical plants, outside of US. In addition to this, India also has a substantial number of plants that adhere to WHO-GMP standards along with those that have received approval from regulatory authorities in various other nations. Indian medicines are preferred worldwide due to their low cost and high quality, earning it the title of "pharmacy of the world."<sup>1</sup>



WHO-GMP – World Health Organisation - Good Manufacturing Practices  
GDP – Gross Domestic Product  
GVA – Gross Value Added  
API – Active Pharma Ingredients

- India plays a vital role in the global vaccine market, as it is one of the largest producers and exporters of low-cost vaccines. According to the World Health Organisation (WHO), India supplies meet 40 to 70 % of the WHO demand for Diphtheria, Tetanus and Pertussis (DPT) and Bacillus Calmette-Guérin (BCG) vaccines, and 90 % of the WHO demand for the measles vaccine.<sup>2</sup>
- The Indian pharmaceutical industry is known for its generic medicines and low-cost vaccines. Globally, India is the largest supplier of generics and is known for its drugs and vaccines.<sup>1</sup> Indian pharmaceutical industry is currently valued at INR3.4 Lakh Crore and Indian pharma products are exported to 200+ countries<sup>1,2</sup>. India supplies over 50% of Africa's requirement for generics, 40% of generic demand in the US, and 25% of all medicine in the UK.<sup>2</sup>
- In terms of value, 97% of the drugs sold in India are generics; but only 10% are unbranded and sold mostly in public health facilities. The rest are *branded generics*, a unique feature of the Indian market.<sup>3</sup> Furthermore, India is the world's largest provider of generic medicines, manufacturing about 60,000 different generic brands across 60 therapeutic areas.<sup>1</sup>

<sup>1</sup> Department of Pharmaceuticals. (2023). *Annual Report 2022-23*.

<sup>2</sup> Invest India. (2022). *Pharmaceuticals*.

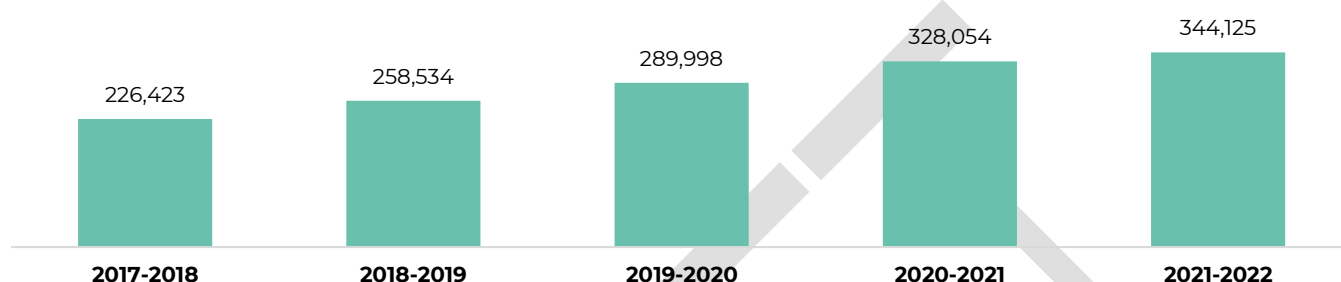
<sup>3</sup> Competition Commission of India. (2021). *Market Study on The Pharmaceutical Sector in India*.

- India also has a strong presence in the active pharmaceutical ingredient (API) industry, with 500 API manufacturers and contributes 8% in the global API industry.<sup>1</sup>

These achievements demonstrate India's significant contribution to the global health sector and its potential for further growth and innovation.

## 1. Pharma Sector Growth<sup>1</sup>

**Fig 1 - Total Indian Pharma Annual Turnover  
(in INR Cr)**

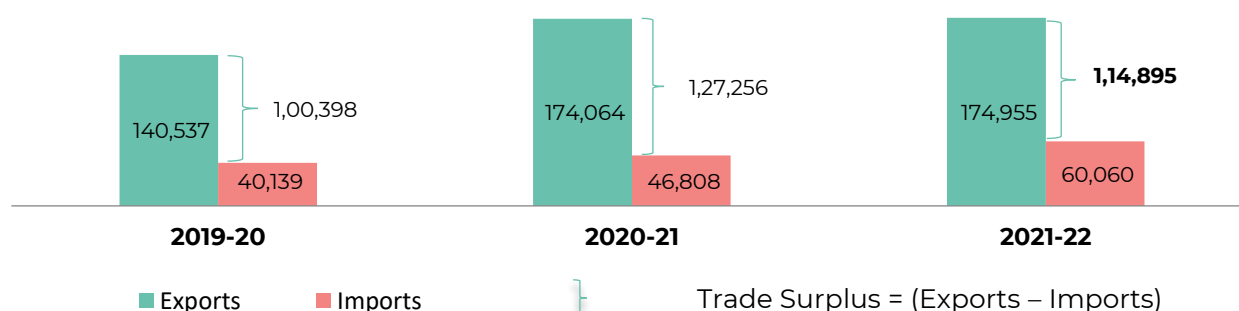


In the FY 2021-2022 the pharma industry showed a 5% growth from the previous year while turnover was INR 3,44,125 Crore.

The Indian pharma industry contributed to 1.32% of Gross Value Added (at 2011-12 constant prices) to the Indian economy in 2021.

## 2. Annual Import-Export of Pharmaceuticals

**Fig 2 - Annual Import-Export  
(in INR Cr)**



Indian pharmaceutical sector plays a significant role in the country's foreign trade and has **consistently maintained a trade surplus**. During 2021-22, a trade surplus of INR1,14,895 Cr was recorded.<sup>1</sup>

Pharma sector contributes to approximately 3.71% of the total FDI inflows into the country, spanning across diverse sectors.<sup>2</sup>

FDI - Foreign Direct Investment

# Sub-Sectoral Trends

## A. Key Pharma Products and Trends

### 1. Key Products<sup>1</sup>

Some of the key products of the Pharmaceutical industry are as below:

|                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Generics</b><br>Generic drugs are unbranded medicines which are equally safe and have the same efficacy as that of branded medicines in terms of their therapeutic value     |  <b>Bulk Drugs</b><br>Bulk drugs, also known as active pharmaceutical ingredients (APIs), are the raw materials used for the production of finished drugs |  <b>Biosimilars</b><br>A biosimilar is that which is similar in terms of quality, safety and efficacy to an approved Reference Biological product based on comparability |
|  <b>OTC Drugs</b><br>OTC drugs, also known as over-the-counter drugs, are medicines that can be sold directly to consumers without a prescription from a healthcare professional |  <b>Vaccines</b><br>Vaccines are biological preparations that provide active immunity against a specific disease or infection                             |  <b>Biologics</b><br>Biologics are medicines that are made from living organisms or their products, such as proteins, antibodies, nucleic acids, cells or tissues        |

### 2. Key Industry Trends

In this section, we will explore some of the key trends that are shaping the future of the pharmaceutical industry in India, such as global vaccine supply, domestic drug research and development, generic drug market growth, cost optimization, and biosimilars and biologics.

**Table 1 - Key Industry Trends**

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Global Vaccine Supplier</b>                                   | <ul style="list-style-type: none"><li>India fulfils a 40-70%, of WHO's requirements for DPT (diphtheria, pertussis or whooping cough, and tetanus) and BCG (Bacille Calmette-guérin) vaccines used to combat tuberculosis.<sup>1</sup></li><li>Telangana's life sciences ecosystem produces one-third of total vaccines produced every year.<sup>4</sup></li></ul>                                             |
| <b>Shifting focus from 'Make in India' to 'Develop in India'</b> | <ul style="list-style-type: none"><li>National Health Policy 2017 has placed priority on promotion of drug research in India.<sup>5</sup></li><li>Boosting domestic production of bulk drugs to reduce imports and self-sufficiency. Financial incentives within the sub-scheme are granted.</li><li>INR 3,000 Cr outlaid for financial assistance to bulk drug parks from 2020 to 2025.<sup>1</sup></li></ul> |
| <b>Healthy market growth of generic drugs</b>                    | <ul style="list-style-type: none"><li>With an extensive production of 60,000 diverse generic brands, India holds 20% of supply share globally.<sup>1</sup></li><li>9,574 Janaushadhi Kendras are functional (as of June 2023) across country providing generic medicines to the masses.<sup>6</sup></li></ul>                                                                                                  |
| <b>Focus on cost optimization</b>                                | <ul style="list-style-type: none"><li>Scheme for Promotion of Bulk Drug and Medical Devices scheme Parks supports in reducing manufacturing cost.<sup>1</sup></li></ul>                                                                                                                                                                                                                                        |

<sup>4</sup> Invest India. (2022). *India's vaccine manufacturing prowess*. Retrieved 1 September 2023.

<sup>5</sup> MOHFW. (2017). *National Health Policy 2017*.

<sup>6</sup> Pharmaceuticals & Medical Devices Bureau of India. (2023). *Pradhan Mantri Bhartiya Janaushadhi Pariojna (PMBJP)*. Retrieved 5 September 2023.

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <ul style="list-style-type: none"> <li>ICMR has established the “Medical Device and Diagnostics Mission Secretariat (MDMS)” to support and catalyse Research and Development of indigenous manufacturing of cost-effective medical devices.<sup>1</sup></li> <li>Updated National List of Essential Medicines released in 2022 has added 36 new items to the list.<sup>7</sup></li> </ul> |
| <b>Biosimilars and Biologicals<sup>10</sup></b> | <ul style="list-style-type: none"> <li>India has 98 biosimilars approved, which is the highest in the world.</li> <li>Biosimilars represents 20% of the total pharmaceutical market in India and are growing at an estimated CAGR of 22% by 2025.</li> </ul>                                                                                                                              |

## Some Prominent manufacturers in the Indian Pharmaceutical Sector

**Table 2 - Prominent manufacturers in the Indian Pharmaceutical Sector**

| Bulk Drugs <sup>8</sup>                   | Biosimilars <sup>9</sup>      | Vaccines <sup>10</sup>       | Generics <sup>11</sup>   |
|-------------------------------------------|-------------------------------|------------------------------|--------------------------|
| Aurobindo Pharma                          | Reliance Life Sciences Group  | Bharat Biotech International | Biocon Biologics         |
| Karnataka Antibiotics and Pharmaceuticals | USV Ltd                       | Biological E Limited         | Piramal Healthcare       |
| Kinvan Pvt.                               | Inbiopro Solutions            | Serum Institute of India     | Mallinckrodt             |
| Natural Biogenex Private                  | BioGenomics                   | Shantha Biotechnics          | Torrent Pharma           |
| Symbiotec Pharmedlab Private              | Biocon Ltd                    | Cadila Healthcare            | Endo International       |
| Macleods Pharmaceutical                   | Zydus Cadila Group            | Ranbaxy Lab                  | Sanofi Pasteur India     |
| Sudarshan Pharma Industries               | Serum Institute of India      | GSK Asia                     | Amneal                   |
| Emmennar Pharma Private                   | Enzene Biosciences Ltd        | Hetero Biopharma Limited     | Glenmark                 |
| Hindys Lab Private                        | Dr. Reddy's Laboratories Ltd  | Indian Immunological Limited | Sawai                    |
| Alkimia Pharma-Chem                       | Chandra Bhagat Pharma Pvt Ltd | Chiron Behring               | Sun Pharma               |
| Meghmani                                  | Lupin Ltd                     | Dano Vaccine and Biological  | Dr. Reddy's Laboratories |
| Sadhana Nitro Chem                        | Cipla Ltd                     | Green signal BioPharma       | Lupin                    |
| Rajasthan Antibiotics                     | RAS Lifesciences Pvt Ltd      | Sanofi Pasteur India         | Hikma                    |
| Centrient Pharmaceuticals India           | Neiss Labs Ltd                | Haffkine                     | Cipla                    |
| Anasia Lab                                | Intas Biopharmaceuticals      | Pasteur institute of India   | Aurobindo Pharma         |

<sup>7</sup> MOHFW. (2022). *National List of Essential Medicines (NLEM), 2022*.

<sup>8</sup> Department of Pharmaceuticals. (n.d.). *List of Applicants KSM*.

<sup>9</sup> BIRAC. (2019). *India: The emerging hub for biologics and biosimilars*.

<sup>10</sup> Department of Pharmaceuticals. (n.d.). *List of Licensed Human Vaccine Manufacturing Facilities in India*. Retrieved 6 September 2023.

<sup>11</sup> Pharmaceuticals & Medical Devices Bureau of India. (2022). *Details of tenders/purchases concluded during the month: 2021:2022*.

## B. Key Sub-Sectors of the Indian Pharma Industry

### 1. Bulk Drugs/API

Active Pharmaceutical Ingredient (API), the biologically active element of a medication responsible for producing the desired medical effect, holds significant importance in the Indian pharmaceutical sector. It constitutes a vital portion of the industry, contributing around 35% of the market.<sup>7</sup>



**3<sup>rd</sup>**

rank globally in  
API production<sup>1</sup>



**500+**

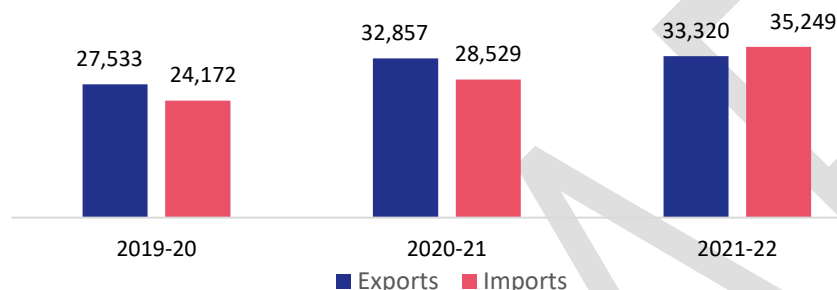
different APIs are  
manufactured in  
India<sup>1</sup>



**57%**

APIs included in  
prequalified  
list of WHO<sup>1</sup>

**Fig 3: API international trade in 2021-2022 (in INR Cr)<sup>14</sup>**



PLI Production Linked Incentive scheme for Bulk Drugs has been implemented with a financial outlay of **INR 6,940 Cr** (For FY 2020-2021 to FY 2029-30) with the aim of enhancing domestic manufacturing of 41 specific critical bulk drugs in India.<sup>13</sup> Currently, a total of 51 out of 249 applications have been approved as part of this scheme.<sup>1</sup>

### Key Government Initiatives for Bulk Drugs & API

The Government of India has introduced several interventions to make India self-reliant in these critical inputs, reduce the dependence on imports, enhance the quality and affordability of medicines, and promote innovation and research in the sector. The key initiatives are:



Production Linked Incentive  
Scheme for Bulk Drugs (PLI 1.0)<sup>1</sup>

India has commenced production of 35 APIs that represent 67% of APIs for which India has 90% import dependency<sup>14</sup>



Bulk Drug Park Schemes

Proposals from Himachal Pradesh, Gujarat, and Andhra Pradesh have received "in-principle" approval for bulk drug park<sup>2</sup>



Foreign Direct Investment

Up to 100% FDI has been allowed through automatic route for Greenfield pharmaceuticals projects<sup>2</sup>

<sup>12</sup> National Informatics Centre. (2022). *Production and Export of Important Medicines*.

<sup>13</sup> Invest India. (2023). *Harnessing India's API Potential*. Retrieved 7 September 2023.

## 2. Vaccines

India holds a prominent position as a global leader in the provision of DPT, BCG, and Measles vaccines. Moreover, India stands as one of the major contributors to the worldwide distribution of affordable vaccines. Every year, India provides 150 Cr vaccine doses, exporting them to more than 150 countries and generating an estimated INR 3,363 Cr in global sales.<sup>15</sup>



**INR 252 Bn**

is the projected valuation of Indian vaccine market by 2025<sup>16</sup>



**40-70%**

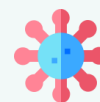
of WHO's demand for DPT, BCG vaccines used to combat TB is fulfilled by India<sup>1</sup>



**220 Cr+**

COVID-19 Vaccination doses distributed<sup>17</sup>

(As of 4 September 2023)



**90%**

of WHO's demand for the measles vaccine is covered by India<sup>1</sup>

### Immunisation<sup>18</sup>

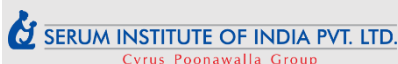
The Universal Immunisation Programme (UIP) of India is one of the largest public health initiatives, with an annual target population of 2.67 crore new-borns and 2.9 crore pregnant women.

Mission Indradhanush launched in Dec 2014, has the aim of full immunisation coverage for 90% of children while focusing on hard-to-reach areas of rural India.

### Covid-19



COVAXIN®, Bharat Biotech's indigenous COVID-19 vaccine, was created in collaboration with the Indian Council of Medical Research (ICMR) - National Institute of Virology (NIV).<sup>19</sup>



is the largest producer and seller of vaccines in the world and are used in 170 countries for their immunisation programmes.<sup>20</sup>

### Vaccine Maitri: India's friendship initiative<sup>21</sup>

India launched the Vaccine Maitri (Vaccine Friendship) initiative - a major diplomatic effort to gift and supply made-in-India Covid-19 vaccines to low-income and developing countries globally following its philosophy of "Vasudaiva Kutumbakam".

A total of **28.1 Cr** doses of Made-in-India COVID-19 vaccines were dispatched to **99** countries along with UN Peacekeepers and UN Health Workers as part of the Vaccine Maitri initiative as of 30 Nov 2022.

Table 3 - Vaccine Maitri

| Country                                                                             |             | Total Supplies in Lakhs<br>(As of June 2023) <sup>20</sup> |
|-------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|
|  | Netherlands | 1,136                                                      |
|  | Australia   | 309                                                        |
|  | Bangladesh  | 281                                                        |
|  | Myanmar     | 212                                                        |
|  | Nigeria     | 98                                                         |

<sup>14</sup> NITI Aayog. (2021). *Investment Opportunities in India's Healthcare Sector*.

<sup>15</sup> Indbiz. (2020). *Ancillary firms look to tap US\$6 billion vaccine business*. Retrieved 4 September 2023.

<sup>16</sup> MyGov. (2023). *India Fights Corona*. Retrieved 4 September 2023.

<sup>17</sup> National Health Mission. (n.d.). *Immunization*. Retrieved 4 September 2023.

<sup>18</sup> Bharat Biotech. (2023). *Home*. Retrieved 4 September 2023.

<sup>19</sup> Serum Institute of India. (n.d.). *About Us*. Retrieved 4 September 2023.

<sup>20</sup> Ministry of External Affairs. (2023). *Vaccine Supply*. Retrieved 12 September 2023.

<sup>21</sup> MEA. (2022). *Annual Report 2022*.

## Highlights



Successful administration of two vaccines, namely Bharat Biotech's Covaxin and Oxford-AstraZeneca's Covishield (manufactured by SII), played a crucial role in protecting the Indian population from the impact of COVID-19.<sup>22</sup>



The Government of India's Aarogya Setu, which was developed through Public Private Partnership to connect health services and the people of India, had augmented the GoI's initiatives in proactively reaching out to and informing users about the potential risk of infection, best practices, and relevant medical advisories pertaining to the containment of the COVID-19 pandemic.<sup>23</sup>



Indian government has introduced the Electronic Vaccine Intelligence Network (eVIN) system, which digitalizes the management of vaccine stock, logistics, and temperature tracking across all levels of vaccine storage, ranging from the national level down to the sub-district level.<sup>18</sup>



National Vaccine Policy of India focuses to strengthen various vaccination programmes to address the issues of vaccine security, vaccine research and development, vaccine quality regulatory system, and product development for public health emergency.<sup>24</sup>



India established National Technical Advisory Group on Immunisation (NTAGI) in 2001 as India's apex advisory body on immunisation. NTAGI provides guidance and advice to the MoHFW on provision of vaccination and immunisation services for the effective control of vaccine preventable diseases in the country.<sup>25</sup>



GAVI (Global Alliance for Vaccine and Immunisation) has partnered with Government of India and will provide 2047 Crore in funding to decrease the number of zero-dose children in India by 30% by 2026 and also aims to build stronger demand for vaccines, strengthen health systems and help introduce the HPV and typhoid vaccines into routine immunisation.<sup>26</sup>

<sup>22</sup> NITI Aayog. (2023). *A New Lens for Innovation in New India*.

<sup>23</sup> National Portal of India. (n.d.). *Fight against COVID-19*. Retrieved 12 September 2023.

<sup>24</sup> MoHFW. (2011). *National Vaccine Policy of India*.

<sup>25</sup> MoHFW. (2015). *Code of Practice NTAGI*.

<sup>26</sup> Gavi. (2023). *Gavi and Government of India establish new partnership to protect millions of children by 2026*. Retrieved 12 September 2023.

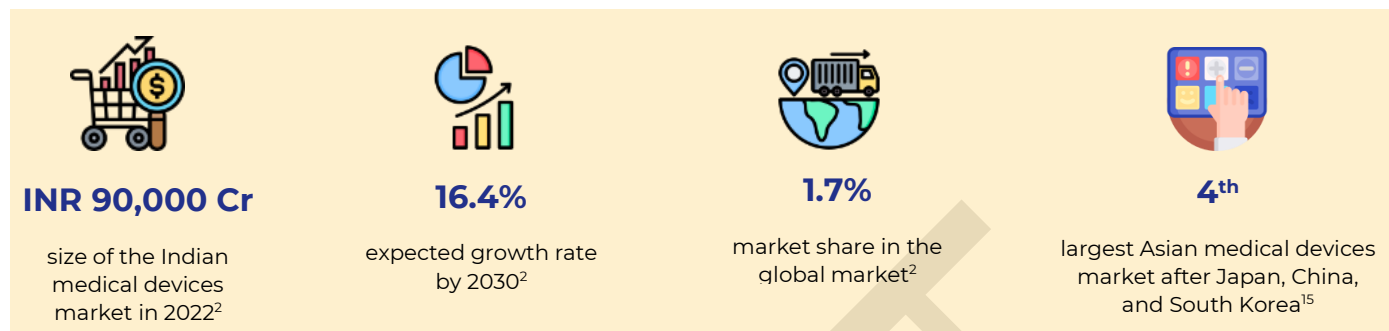
## List of vaccines manufactured in India<sup>11</sup>

Table 4 - List of vaccines manufactured in India

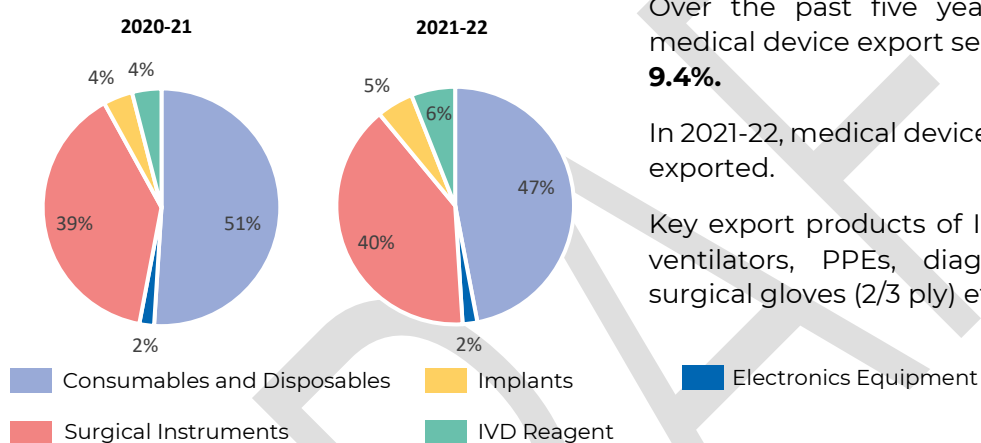
| Type of Manufacturer                                               | Name of Vaccine             |                              |                              |                                                 |                                   |
|--------------------------------------------------------------------|-----------------------------|------------------------------|------------------------------|-------------------------------------------------|-----------------------------------|
| <b>Indigenous Manufacturers in Government</b>                      | BCG                         | Tuberculine                  | DTP                          | Yellow fever                                    | JE                                |
|                                                                    | TT                          | DT                           | Concentrated DTP             | Inactivated Rabies vaccine                      |                                   |
| <b>Indigenous Manufacturers in Public Sector Undertaking (PSU)</b> | bOPV                        | mOPV                         | Rabies                       | DTP                                             | TT                                |
|                                                                    | DT                          | Hep- B                       | DTwP- HepB- Hib              |                                                 |                                   |
| <b>Indigenous Manufacturers in the Private Sector</b>              | Hib                         | Rabies                       | bOPV                         | mOPV                                            | DTP+Hib+HepB                      |
|                                                                    | Vi polysaccharide Typhoid   | H1N1                         | DTP                          | DTP+HepB                                        | Rotavirus vaccine                 |
|                                                                    | Inactivated JE vaccine      | Typhoid+TT Conjugate Vaccine | DTP+Hep-B+Hib                | DTP+Hib                                         | TT                                |
|                                                                    | JE                          | Hep-B                        | Diphtheria                   | Pertussis                                       | Hib                               |
|                                                                    | DT                          | IPV                          | Meningococcal Polysaccharide | bOPV                                            | Vi Polysaccharide Typhoid Vaccine |
|                                                                    | Meningococcal               | Typhoid                      | Tetavalent Influenza         | Pandemic influenza H1N1 2009 monovalent vaccine | BCG                               |
|                                                                    | DPT+Hep-B+Hib+IPV           | Diphtheria                   | WP                           | Typhoid polysaccharide                          | Hib Conjugate vaccine             |
|                                                                    | MMR                         | Measles                      | Rubella                      | Meningococcal A conjugate                       | Mumps                             |
|                                                                    | Measles +mumps              | Measles+ Rubella             | DTaP                         | DPT+Hep B                                       | Oral cholera vaccine              |
|                                                                    | Pneumococcal Polysaccharide | Haemophilus influenza        | Yellow fever                 | Seasonal influenza                              | Varicella                         |

### 3. Medical Devices

The Medical Devices sector has been recognized as a sunrise sector under the Make in India campaign. Backed by the growing medical needs of the country, Indian medical device market is expected to quadruple by 2030.<sup>27</sup>



**Fig 4: Category-wise export data<sup>1</sup>**

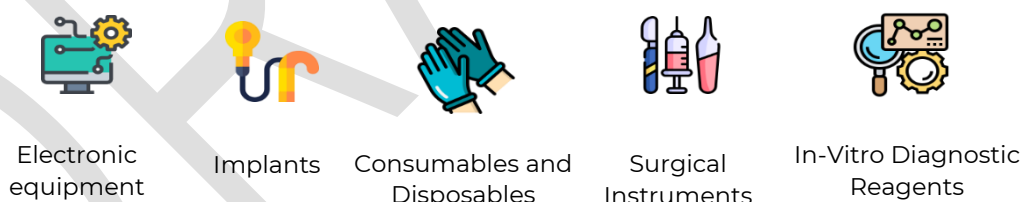


Over the past five years (2017-18 to 2021-22), the medical device export sector has witnessed a CAGR of **9.4%**.

In 2021-22, medical devices worth **INR 23,982.3 Cr** were exported.

Key export products of India in the segment include ventilators, PPEs, diagnostic kits, sanitizers and surgical gloves (2/3 ply) etc.

#### Medical Device Fields



### National Medical Device Policy, 2023

The vision of the policy is to pursue an accelerated growth trajectory with a patient-centric focus, aiming to establish India as a global leader in medical device manufacturing and innovation.<sup>28</sup>

The goal is to attain a 10-12% share in the expanding global market within the next 25 years.

The policy is anticipated to facilitate substantial growth in the Medical Devices Sector, aiming to increase its current value of INR 90,152 Cr to INR 4,09,783 Cr within the next five years.

Strategies: Regulatory Streamlining, Enabling Infrastructure, Facilitating Research and Development and Innovation, Attracting Investments in the Sector, Human Resources Development and Brand Positioning and Awareness Creation.

4

**Medical Device Parks:** Himachal Pradesh, Tamil Nadu, Madhya Pradesh & Uttar Pradesh

<sup>27</sup> Invest India. (n.d.). *Medical Devices*. Retrieved 12 September 2023.

<sup>28</sup> Department of Pharmaceuticals. (2023). *National Medical Device Policy 2023*.

## C. Pharmaceutical Price Fixation

NPPA was established with the objective of setting and revising prices for controlled bulk drugs and formulations. It is also responsible for ensuring the availability of medicines at regulated prices operating under the Drugs (Prices Control) Second Amendment Order, 2023<sup>29,30</sup>. While approximately 17.7 % of India's pharmaceuticals market (in terms of value) is subject to price regulation, competition is expected to be the primary source of price discipline for the remainder of the market.<sup>5</sup>

### Principles for Drug Pricing<sup>29</sup>



Essentiality of  
Drugs

**INR 3,532 Cr**

Estimated annual savings (FY April 2022 to March 2023) in moving annual turnover (MAT) for 651 formulations fixed under NLEM 2022



Control Formulations  
Prices



Market Based  
Pricing

**6.4%**

Average reduction in the ceiling price of scheduled drugs under NLEM 2022

### Ceiling Price<sup>1</sup>

Until March 2023, NPPA has determined the maximum prices for 651 formulations

29 therapeutic categories have been created for medicines covering 384 drugs under NLEM 2022

### Retail Price<sup>1</sup>

Manufacturing companies submit Form-I applications based on which retail price is fixed by NPPA

2,299 'new drugs' retail prices were notified by NPPA till March, 2023

### Pricing of Medical Devices Updates<sup>1</sup>

Coronary stents ceiling prices were notified in 2017 and included in schedule-I in 2016

Ceiling price of Orthopaedic Knee Implants, a non-scheduled medical device was included in 2017

### Pharma Jan Samadhan<sup>31</sup>

It is part of the Digital India initiative and offers consumers an efficient and prompt mechanism for resolving grievances, ensuring an effective resolution system for complaints regarding medicine pricing, shortages, and unavailability.

NPPA: National Pharmaceutical Pricing Authority

NLEM: National List of Essential Medicines

<sup>29</sup> Department of Pharmaceuticals. (2023). *National Pharmaceutical Pricing Authority*. Retrieved 14 September 2023.

<sup>30</sup> Ministry of Chemicals and Fertilizers. (2023). *Drugs (Prices Control) Second Amendment Order, 2023*.

<sup>31</sup> National Pharmaceutical Pricing Authority. (2018). *Pharma Jan Samadhan*. Retrieved 14 September 2023.

# Opportunities and Challenges

## A. Opportunities

### Emerging Hub for Clinical Trials<sup>2</sup>

Cost of conducting trials in India is ~40-60% less than in developed markets. With 500+ contract research laboratories, it offers ample resources for conducting trials. Diverse patient population and rising expenses in developed markets is making India a hub for clinical trials.

### Biosimilar Market Growth<sup>32</sup>

Products, with a market size of INR 7,541 Cr in 2016. The Indian market for biosimilars is expected to reach INR 18,033 Cr by 2025.

### Boost in Pharma Production<sup>2</sup>

There is an opportunity to enhance the domestic production of pharmaceuticals with the aid of the recent PLI schemes. This, in turn, opens up investment opportunities in other pharma segments as well.

### Surge in Tech-Based Manufacturing

Medical device players can expect new opportunities to arise in key segments such as the development of tools that streamline emergency care and enhance medical infrastructure by leveraging technology-driven optimization.<sup>16</sup> Focus on future ready technologies like Biologics, continuous flow technology, use of AI and automation in manufacturing will further propel India.<sup>33</sup>

## B. Challenges

### Less domestic production of medical devices<sup>1</sup>



Less domestic production and more imports results in limited availability of medical devices, particularly in remote areas.

In 2021-2022, **medical devices** worth only INR 24,038 Cr were exported, and INR 70,223 Cr had to be imported.

### High reliance for API on China



**70% of the bulk drugs/API** are **imported from China** (example-Heparin).<sup>2</sup>

Heavy reliance on a single country makes the industry vulnerable to disruptions in the supply chain (ex - impacts the manufacturing of essential medicines).

India imported a total of **INR 35,249 Cr worth of APIs and Bulk drugs** in 2021-22 of which INR 23,273 Cr was from China.<sup>15</sup>

<sup>32</sup> Invest India. (2020). *Biotechnology*.

<sup>33</sup> Department of Pharmaceuticals. (2021). *Draft Policy to Catalyse Research & Development and Innovation in the Pharma- MedTech Sector in India*.

## Falling behind in value of drug<sup>34,35</sup>



India ranks **14<sup>th</sup>** in the world in the pharma industry in terms of value despite being the **3<sup>rd</sup>** biggest manufacturer.

The government has recognized the need to move up the value chain and is focusing on both innovation and manufacturing by moving India beyond generics.

## Awareness



**Cost savings from generic drugs is not much known** to the general public and physicians.

Patients may have concerns about the quality, safety, or efficacy of generic drugs compared to branded counterparts.<sup>36</sup>

## High out of Pocket Expenses



Households **out of pocket expenditure on health is 60% of the total health expenditure**. This high burden results in non-adherence to prescribed medicines.<sup>35</sup>

## Issue of Counterfeit Medicines



WHO's staggering statistic reveals nearly **10.5% of medications globally are of counterfeit quality**.<sup>37</sup>

India still struggles with the problem of counterfeit medicine with **53% share by value** in Global Custom Seizures.<sup>38</sup>

## Supply Chain Issues



Margin and profitability are being squeezed by the escalating pricing pressure in the regulated market. Key factors include a delay in new launches, increased competition in commoditized, simple-to-manufacture items, and customer consolidation.<sup>39</sup>

India continues to rely on imports of essential starting materials, intermediates, and APIs from China, with the proportion of reliance growing over time, thereby exposing India to **raw material supply interruptions and pricing volatility**.<sup>39</sup>

## Prime Minister Jan Aushadhi Pariyojna Issues<sup>40</sup>



The Jan Aushadhi Kendra's (JAK's) have limited variety of therapeutic groups and dosages thus hindering their use by the public.

Reluctance of medical professionals to prescribe generic drugs is another major issue.

Other issues include lack of supply chain monitoring and permission issues from respective state governments for opening new stores.

<sup>34</sup> WIPO. (2022). *Global Innovation Index 2022*.

<sup>35</sup> MOHFW. (2018). *National Health Accounts Estimates for India*.

<sup>36</sup> EY. (2021). *Indian Pharmaceutical Industry 2021: Future Is Now*.

<sup>37</sup> Pathak, R., Gaur, V., Himanshu Sankritayan, & Jaideep Gogtay. (2023). *Tackling Counterfeit Drugs: The Challenges and Possibilities*.

<sup>38</sup> OECD/EUIPO. (2020). *Trade in Counterfeit Pharmaceutical Products*.

<sup>39</sup> Invest India. (2018). *Trends and Opportunities for Indian Pharma*.

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# Cross Cutting Themes

## Climate



**Significance:** The industry needs to adopt sustainable practices and technologies to reduce its environmental impact and mitigate the risks of climate change.

**Current efforts:** Some important steps by the government of India and the industry to support climate action in the pharmaceutical sector include the NAPCC, the PLI Scheme, and the voluntary adoption of green standards by some leading companies.

**Way Forward:** Strengthen the collaboration, coordination, capacity, and awareness among various stakeholders, and to leverage the potential of digital technologies for climate adaptation and mitigation.

## Education



**Significance:** There is a shortage and uneven distribution of qualified pharmacists, a huge skill gap among the graduates, and the low level of infrastructure and facilities for pharmaceutical education and research.

**Current efforts:** The Government of India and the industry have taken some steps to promote and facilitate pharmaceutical education and research, such as the establishment and expansion of NIPERs, the introduction and revision of various courses and programmes in pharmacy education, and the promotion and support of industry-institution collaboration.

**Way Forward:** There is a need to increase the investment and allocation of resources for pharmaceutical education and research, to strengthen the linkages and partnerships among various stakeholders, and to enhance the accessibility and affordability of pharmaceutical education for all sections of society.

## Research and Development



**Significance:** While India has worked to reduce cost of manufacturing pharmaceuticals in India, the value of these remains relatively low. Local research and development of drugs is essential to develop high value pharmaceuticals.

**Current efforts:** The Government of India and the industry have taken some steps to support and incentivize R&D in the pharmaceutical sector, such as the New Drugs and Clinical Trials Rules, 2019, the National Biopharma Mission, BIRAC, etc., the establishment and strengthening of various institutions and platforms for R&D collaboration and coordination, such as ICMR, CSIR, DBT, etc.

**Way Forward:** Increase the investment and allocation of resources for R&D in the pharmaceutical sector, to strengthen the linkages and partnerships among various stakeholders, and to enhance the regulatory and intellectual property framework and environment for R&D in the pharmaceutical sector.

NAPCC: National Action Plan on Climate Change

PLI Scheme: Production Linked Incentive Scheme

NIPER: National Institute of Pharmaceutical Education and Research

BIRAC: Biotechnology Industry Research Assistance Council

ICMR: Indian Council of Medical Research

CSIR: Council of Scientific and Industrial Research

DBT: Department of Biotechnology

## Digital Technology



**Significance:** Digital technology can enhance the productivity, efficiency, quality, innovation, and competitiveness of the pharmaceutical industry

**Current efforts:** Government has launched various initiatives like digital India and the establishment and strengthening of various institutions and platforms for digital technology collaboration and coordination, such as NIC, NeGD, C-DAC, etc.

**Way Forward:** Adoption and application of advanced technologies, such as AI, ML, big data analytics, IoT, blockchain, etc., to enhance the efficiency, quality, speed, and accuracy of various processes and functions in the pharmaceutical sector.

## Financial Affordability



**Significance:** Challenges in terms of financial affordability include low penetration and utilization of health insurance and other risk protection mechanisms among the population, the high out-of-pocket expenditure on health care by the consumers, and the lack of access to adequate and timely credit and finance by the industry players, especially the small and medium enterprises (SMEs).

**Current efforts:** Some prominent efforts include the launch of Pradhan Mantri Bhartiya Janaushadhi Pariyojana for distribution of essential medicines at economical prices.

**Way Forward:** Enhance the regulatory and policy framework and environment for financial inclusion in the pharmaceutical sector.

NIC: National Informatics Centre

NeGD: National eGovernance Division

C-DAC: Centre for Development of Advanced Computing

# Ministries and Departments

In Indian governance, policy convergence plays a critical role. It involves the coordination and collaboration of different ministries and departments to attain shared goals and objectives.

**Table 5: Ministries and Departments**

| <br><b>Ministry</b>                     | <br><b>Department / Organisations</b>                                    | <br><b>Functions and Roles</b>                                                                   |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Ministry of Chemicals and Fertilizers   |  <b>Department of Pharmaceuticals</b>                                    | Nodal Department overseeing overall sector development and to regulate various complex issues related to pricing and availability of medicines                                      |
|                                                                                                                          |  <b>National Pharmaceutical Pricing Authority (NPPA)</b>                 | Independent Regulator for pricing of drugs that works to ensure availability and accessibility of medicines at affordable price                                                     |
|  Ministry of Health and Family Welfare |  <b>Central Drugs Standard Control Organisation (CDSCO)</b>             | Responsible for drug standard control and grant of licenses for special category drugs                                                                                              |
|                                                                                                                          |  <b>Indian Council of Medical Research (ICMR)</b>                      | Apex body in India for the formulation, coordination and promotion of biomedical research                                                                                           |
|  Ministry of Science and Technology    |  <b>Department of Biotechnology (DBT)</b>                              | Nodal agency supporting research and its applications in the Life Sciences and promotes and accelerates the pace of development and large-scale use of biotechnology in the country |
|                                                                                                                          |  <b>Council of Scientific and Industrial Research (CSIR)</b>           | Provides significant technological intervention in many areas concerning societal efforts including chemicals, drugs, genomics and biotechnology                                    |
|  Ministry of Commerce and Industry     | <br><b>Pharmaceuticals Export Promotion Council of India</b>           | Trade council that works on overall promotion and support of pharmaceutical sector                                                                                                  |
|  Ministry of AYUSH                     | <br><b>Pharmacopoeia Commission for Indian Medicine and Homeopathy</b> | PCIMandH acts as a Central Drug Testing cum Appellate Laboratory for Indian system of Medicine and Homoeopathy.                                                                     |

# Government Interventions

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## National Cold Chain and Management Information System (NCCMIS)<sup>41</sup>

The National Cold Chain and Management Information System (NCCMIS) serves as a repository for cold chain equipment as well as a tool for planning and future cold chain and vaccine management innovations.

84 thousand electrical and 1.2 million non-electrical cold chain equipment are covered under this system.

Used by both Centre and State for

- Policy level decisions
- Resource planning of cold chain points, rational distribution of Cold Chain Equipment and capacity building of human resources
- Monitoring of key cold chain performance indicators



## Drug and Vaccine Distribution Management System (DVDMS)<sup>42</sup>

DVDMS is a system that manages the purchase order, distribution, quality control, and monitoring of drugs and vaccines for various disease control programmes. It aims to provide quality drugs to the hospitals at a reasonable price and to ensure timely and transparent delivery to the beneficiaries by enabling data based informed decisions.

Along with central dashboard, many states have already implemented DVDMS for distribution of drugs and vaccines to the District Drug Warehouses (DWH), District Hospitals (DH), Community Health Centres (CHC) and Primary Health Centres (PHC).

Some Features of DVDMS include:

- Easy to read UI
- Role based User access
- Filters
- Report Drill Down
- Comparison

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<sup>41</sup> National Cold Chain and Vaccine Management Resource Centre. (n.d.). *NCCMIS Web Application*. Retrieved 18 September 2023.

<sup>42</sup> MoHFW. (n.d.). *DVDMS Central Dashboard*. Retrieved 1 October 2023.

e-Aushadhi streamlines the drug and vaccine supply chain by automating the tasks of procurement, inventory, distribution, and quality assurance. The system is used by Central Medical Services Society and some states like Rajasthan, Jharkhand, etc.

It aims improve the efficiency and effectiveness of the drug warehouse system and provides cost-effective and timely medicines to the people.

e-Aushadhi system helps with:

- Procurement
- Statistical Data Management
- Efficient flow of medicines
- Streamlining of quality management process
- Financial Management
- Alert management system
- Online payments

## Central Government Schemes

### A. Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)<sup>8</sup>

- Launched in 2008
- Dedicated Outlets: Janaushadhi Kendras
- Product Basket: 1,800 drugs and 285 surgical items
- Jan Aushadhi Suvidha Napkins: INR 1 per pad
- Financial Outlay: INR 490 Cr from 2020 till 2025
- Since beginning, scheme has led to savings of INR ~18,000 Crore for citizens
- Average sales per Kendra per month: INR 1,05,000

### B. Scheme for Strengthening of Pharmaceuticals Industry (SPI)<sup>1</sup>

- Assistance to Pharmaceutical Industry for Common Facilities
- Pharmaceutical Technology Upgradation Assistance Scheme
- Pharmaceutical and Medical Devices Promotion and Development Scheme
- Scheme for Promotion of Bulk Drug Parks
- Promotion of Medical Devices Parks

### C. Production Linked Incentive (PLI) Schemes<sup>1</sup>



**API**

Approved in 2020 to reduce import burden and promote self-reliance

Financial outlay: INR 6,940 Crore from 2020 till 2030

51 applications approved out of 249

<sup>43</sup> Central Medical Services Society. (n.d.). *E-aushadhi*. Retrieved 18 September 2023.



## Medical Devices

Approved in 2020, it intends to attract investments in the medical device sector

Financial outlay: INR 3,420 Cr from 2020 till 2028

21 applications approved across cancer care medical devices, radiology, etc.



## Pharmaceutical

Approved in 2021, to enhance production capabilities through product diversification

Financial outlay: INR 15,000 Cr 2020 till 2029

Biopharmaceuticals, API not covered in other scheme, repurposed drugs

### D. Scheme of Consumer Awareness, Publicity and Price Monitoring (CAPPM)<sup>44</sup>

Aims to regulate the prices of medicines, monitor their availability and create public awareness through NPPA and PMRUs.

Price Monitoring and Resource Units (PMRUs) are registered societies at the State/UT level that provide technical assistance to the State Drug Controllers and NPPA for data collection, analysis, enforcement and consumer education.

PMRUs undertake various activities such as market-based data collection, monitoring of price movement, collection of test samples, advertisement and publication, training and workshops, etc.

<sup>44</sup> Department of Pharmaceuticals. (n.d.). *Consumer Awareness, Publicity and Price Monitoring (CAPPM)*.

# Way Forward



## Research and Development<sup>1</sup>

The industry is focusing on advancing in the competitive markets of generic and specialty medicine. The recently announced Centers of Excellence in the 2023-24 budget will foster pharmaceutical research and promote innovation.



## Boost Domestic Manufacturing and Supply Chain<sup>28</sup>

Recent government schemes, such as PLI schemes for domestic manufacturing of API, medical devices, pharmaceuticals and scheme for promotion of bulk drug parks. These will help boost local production, decrease dependency on imports and will



## MedTech Expo 2023<sup>1</sup>

Department of Pharmaceuticals has launched a distinctive platform, starting in 2023, to showcase the strengths and capabilities of the Indian medical devices' ecosystem. To foster collaboration, networking, and bring together diverse stakeholders.



## Focus on Generic and Biosimilar Markets<sup>36</sup>

Continue to leverage India's strength in generic drug manufacturing to provide affordable healthcare solutions globally. Additionally, invest in the development and production of biosimilars, which offer significant growth potential.



## Increased FDI Flows<sup>1</sup>

Investment-friendly FDI policies are in place to promote foreign inflows in the sector. Medical devices are allowed a 100% of inflows, in pharmaceuticals, up to 100% are allowed in greenfield projects and up to 74% in brownfield projects under automatic route.



## Patent Expiration<sup>13</sup>

Globally, between 2018-2024, patents worth INR306 Cr will expire, presenting a highly profitable opportunity for India's pharmaceutical market. Not only generic drugs, but other segments too have opportunity now.

# Best Practices

## 1. Hi-Tech Laboratory under NUHM in Bengaluru a Concept of Hub and Spoke Model, Karnataka<sup>45</sup>



### Background

- The problem is the high out-of-pocket expenditure for diagnostics by urban poor, as per NSSO-2016.
- The limited range of diagnostics available at UPHCs (Urban Primary Health Centres) under NUHM adds to the burden on tertiary care centre and patients.



### Steps taken by the Government

- 6 Hi-Tech Laboratories were established in Government Hospitals (UCHC) of ULBs under NUHM in Bruhat Bengaluru Mahanagara Palike (BBMP) in April 2018.
- Hi-Tech Laboratories operate as hubs under “Hub and Spike Model” design, with Special Outreach Camps and UPHCs as spokes.
- Laboratory, Radiology, Mammography and Ultrasonography services are provided free of cost for urban poor.
- 83 types of laboratory investigations at the hub and 19 types at the spokes (UPHCs).
- Basic lab services are provided in UPHCs and advanced diagnostics are referred/ transported to Hi-Tech Laboratory.
- Samples of active screening and referrals from Health Kiosk are transported to Hi-Tech Laboratories.
- Hi-Tech Lab services are used in mass screening camps of NCD, Active Case Finding campaigns for X-ray, LCDC (leprosy case detection campaign), etc in urban areas.
- Dedicated software for enrolment and reporting the results to the patient through message and to UPHC/UCHC.
- Five specialist doctors are supported by 30 paramedical staff in these laboratories through NUHM.
- An incentive of Rs. 200 is paid to the lab technician/ Group-D for the transport of samples from spokes to hub initially. State will propose for courier system in FY 2020-21 PIP.



### Impact generated

- On average, 100-120 patients seek services at the Hi-Tech laboratories.
- The total tested samples have increased by 200 times in 2018-19 compared to previous year.



<sup>45</sup> NHSRC. (2019). *We care... Good, Replicable and Innovative Practices*.



## 2. Mukhyamantri Nihshulk Dava Yojna, Rajasthan<sup>46</sup>



### Background

The Chief Minister's Drug Scheme is a scheme that provides free medicines to all patients coming to government hospitals in Rajasthan. The scheme was started on 2 October 2011 and covers medicines included in the essential drug list.



### Steps taken by the Government

- The government has set up 40 district drug warehouses at the district headquarters to distribute medicines to all medical institutions.
- The government has also increased the number of medicines, surgical and sutures in the essential drug list from 832 items in 2018-19 to 2,707 items in 2023.
- The number of items has also been increased for different levels of health facilities, such as medical colleges, district hospitals, community health centres, primary health centres and sub-centres.



### Impact generated

- The scheme has improved the availability and accessibility of quality medicines for the people of Rajasthan.
- It has also reduced the out-of-pocket expenditure of patients and enhanced their satisfaction with the public health system.
- The scheme has also contributed to the prevention and treatment of various diseases and improved the health outcomes of the population.

<sup>46</sup> Government of Rajasthan. (n.d.). *Mukhya Mantri Nishulk Dava Yojna*. Retrieved 15 September 2023.

### 3. Initiatives to Reduce 'Out of Pocket' Expenditure of Patients Through Revamped Medical Store Procurement, West Bengal<sup>47</sup>



#### Background

The case study is about the efforts of the West Bengal Government to reduce the out-of-pocket expenditure of patients who rely on the public health care system.

The case study covers three efforts: improving the free distribution scheme of medicines and consumables, setting up fair price medicine shops under PPP mode, and promoting prescription of medicines in generic names by doctors of government hospitals.



#### Steps taken by the Government

- For the free distribution scheme, the government increased the budget, introduced e-procurement, empanelled drug testing labs, installed web enabled software for store management, created a vendor portal, and integrated the scheme with NHM.
- For the fair price medicine shops, the government selected private partners based on the maximum discount offered, ensured quality assurance by incorporating quality parameters and conducting quality audits, and set up a monitoring mechanism by installing software and forming committees.
- For the prescription of medicines in generic names, the government issued order to instruct doctors to do so and initiated prescription audit to ascertain rational use of drugs.



#### Impact generated

- The efforts have shown positive outcomes in terms of benefits to the patients.
- The efforts have increased the availability, quality, and confidence of patients and doctors on drugs under free scheme.
- The efforts have reduced the dependency of patients on private retail pharmacies that charge higher prices for branded medicines.
- The efforts have also increased the prescriptions in generic names by doctors especially in last two to three months.

The screenshot displays the Vendor Portal for the Health and Family Welfare Department, Govt. of West Bengal. It shows a 'Bill Status Report' table with columns for S.No., District Name, Order No., Order Date, Order Value, Bill Value, Status, and Vendor Remarks. The table lists several orders for various districts including Bardhaman, Bankura, and Burdwan, with statuses ranging from 'Cheque ready' to 'RECEIVED PAYMENT'.

| S.No. | District Name | Order No.              | Order Date | Order Value | Bill Value | Status           | Vendor Remarks   |
|-------|---------------|------------------------|------------|-------------|------------|------------------|------------------|
| 1     | Bardhaman     | 2013-2013/02/06/DC079  | 18/05/2012 | 17798.80    | 18413.00   | Cheque ready     | Received payment |
| 2     | Bardhaman     | 2013-2013/02/06/DC089  | 18/05/2012 | 73942.47    | 74833.00   | Cheque not recd. | Nil              |
| 3     | Bardhaman     | 2013-2013/02/06/DC087  | 18/05/2012 | 16596.56    | 16513.00   | Cheque ready     | Received payment |
| 4     | Bardhaman     | 2013-2013/02/06/DC066  | 18/05/2012 | 16246.22    | 17071.00   | Cheque ready     | Nil              |
| 5     | Bardhaman     | 2013-2013/02/06/DC057  | 13/06/2012 | 124626.75   | 123627.00  | RECEIVED PAYMENT | Nil              |
| 6     | Bardhaman     | 2013-2013/02/06/DC038  | 14/09/2012 | 18413.30    | 18413.00   | Cheque ready     | Nil              |
| 7     | Bardhaman     | 2013-2013/02/06/DC026  | 18/05/2012 | 150605.91   | 138625.00  | Received Payment | Update Cancel    |
| 8     | Bardhaman     | 2012-2012/02/06/DC0146 | 19/01/2013 | 150605.91   | 138625.00  | Cheque ready     | Nil              |
| 9     | Bardhaman     | 2013-2013/02/06/DC084  | 25/01/2012 | 117786.13   | 127786.00  | TO FORM          | Consolidated     |
| 10    | Bardhaman     | 2013-2013/02/06/DC057  | 13/06/2012 | 124626.75   | 124627.00  | RECEIVED PAYMENT | Nil              |

<sup>47</sup> NHSRC. (2021). *West Bengal Drugs Procurement Policy*.



# Recommended Readings

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